

Krishna Badikela

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Education

The Ohio State University

M.S. in Computer Science & Engineering

Columbus, OH

Aug 2024 – May 2026

SRM Institute of Science and Technology

B.Tech. in Computer Science & Engineering, Specialization in Big Data Analytics

Kattankulathur, India

Jul 2020 – May 2024

Publications

One Form to Transfer Them All: Pretraining Multilingual Language Models Beyond Native Orthography

Muge Zhang, Aaron Jencks, **Krishna Badikela**, Yulia Tsvetkov, Sachin Kumar

EMNLP 2026

Experience

Irys Technologies

AI Engineering Intern

San Francisco, CA

Dec 2025 – May 2026

- Built claude-code-machine, a production agentic middleware server ingesting webhook events from 6 sources (Notion, Retell, n8n, Slack, GitHub) across 20+ REST API endpoints, routing each through a 3-layer deterministic/LLM/fallback pipeline into Claude Agent SDK execution
- Designed a multi-provider LLM classification layer (Gemini, Claude Haiku, LiteLLM) with in-memory TTL+LRU response caching and exponential backoff retry; applied LLM inference optimization techniques cutting redundant call latency by ~95% on cache hits
- Automated CalSTRS F496 payroll compliance validation via end-to-end LLM integration using DSPy and Qwen, reducing per-cycle review time from 4 hours to under 3 minutes for a pension fund managing \$340B+ in assets

The Ohio State University — NLP Research Lab

Student Research Engineer

Columbus, OH

Apr 2025 – May 2026

- Pretrained GPT-2 models from scratch at 4 scales (467M to 1.44B params) across 8 languages and 4 writing systems using mixed-precision distributed training on 8x NVIDIA H100 GPUs
- Engineered a self-rescheduling SLURM pipeline managing 500K+ training iterations with automatic checkpoint recovery; developed 12 custom BPE tokenizers from a 137GB Hindi-Urdu corpus
- Validated IPA achieves +5.1 F1 on Polish XNLI and +4.6 on MASSIVE over text baselines; romanization enabled zero-shot Bengali transfer from 3.56 to 31.24 F1; co-authoring EMNLP 2026 submission

The Ohio State University — Fisher College of Business

Graduate Research Associate

Columbus, OH

May 2025 – May 2026

- Constructed a GPT-4o classification pipeline for automotive investment event detection across 45 firms; lifted accuracy from 78% to 93% via few-shot prompting and chain-of-thought reasoning
- Fine-tuned FinanceBERT on 5,000+ firm reports for aspect-based sentiment analysis; achieved 18% precision improvement; deployed async LiteLLM inference pipeline over 1,081 documents with 98.8% run reliability

Deepija Telecom Pvt. Ltd

Machine Learning Intern

Hyderabad, India

Jul 2023 – Sep 2023

- Architected and deployed a production RAG pipeline (LangChain + FAISS) indexing 1M+ knowledge triples; served via FastAPI, cutting query latency 80% and reducing support ticket volume 22% post-launch

Projects

dhvani — Hinglish Normalization Library [PyPI / GitHub]

- Published Python package to PyPI normalizing Hinglish text using IPA as a phonetic bridge between Devanagari and Latin scripts; supports `pip install dhvani` for code-switched Hindi-English normalization in a single call

SMM4H 2024: Multilingual Adverse Drug Event Detection [GitHub]

- Secured 3rd place overall; trained XLM-RoBERTa on <1% positive-class data across 4 languages; reached 78.6% F1-macro on English and $\approx 9\times$ F1 improvement on Russian via SMOTE and language-specific threshold tuning

Technical Skills

Modeling

PyTorch, Hugging Face Transformers, XLM-RoBERTa, spaCy, BPE Tokenization, Cross-lingual Transfer, IPA Phonemization

LLM / RAG

LangChain, FAISS, RAG Pipeline, Prompt Engineering, Few-shot/Chain-of-Thought, LLM Fine-tuning, LiteLLM, GPT-4o, OpenAI API, Model Deployment, Inference Optimization

Infra / MLOps

SLURM/HPC, AWS (SageMaker, S3), Docker, WandB, Git, Linux, Distributed Training

Backend / Web

FastAPI, REST APIs, PostgreSQL, DSPy, Qwen

Languages

Python, SQL, Java, C++